and the slope of the valley. Sometimes, the surface of the glacier forms cracks known as **Crevasses**. Crevasses are the deep fissure of variable width in the surface of a glacier. These crevasses are dangerous to the Mountaineers.



Figure 4.16 crevasses

4.7.3 Action of glaciers

The glacier performs three actions namely erosion, transportation and deposition.

A glacier erodes its bedrock by the action of (1) Plucking and (2)Abrasion

(1) Plucking

The glacier plucks big pieces of rocks from the valley floor and creates large grooves or hollows. These pieces are dragged along the valley floor as the glacier moves. The boulders and rocky floor are grounded by mutual contact.

(2) Abrasion

Pure ice is capable of wearing down massive rocks when equipped with angular rock fragments. The glacier can groove, scratch, and chisel the rock surface. It has a powerful abrasive effect.

As a result, a glacier during its lifetime creates various landforms which may be classified into erosional and depositional landforms.

4.7.4 Erosional landforms of Glaciers

The landforms created by glaciers are mainly found in the mountainous regions. The chief erosional landforms by the glaciers are as follow:

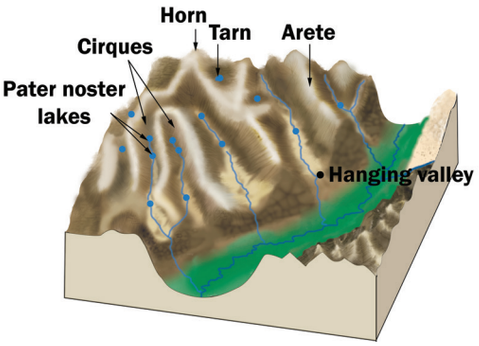


Figure 4. 17: Erosional landforms of Glaciers

1. U-Shaped Valley

U-Shaped Valley is a typical glacial feature. Since glacial mass is heavy and slow moving, erosional activity is uniform in all directions. A steep-sided curved bottom valley has a U shaped profile.

2. Hanging Valley

Hanging valley is formed when tributary glaciers are unable to cut as deeply as main ones and remain "hanging" at higher levels than the main valley as discordant tributaries. These tributary valleys appear hanging over the main valley and enter the main valley at some height.

3. Cirque and Tarn

A Cirque or Corrie is an amphitheater-shaped hollow basin cut into a mountain ridge. It has a steep-sided slope on three sides, an open end on one side and a flat bottom. When the ice melts, the Cirque may develop into a Tarn Lake and the whole thing appears like a big armchair.

4. Aretes

It is a steep-sided, sharp-tipped saw toothed ridges which have undergone glacial erosion from two sides. These comb like ridges are called as arete.

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